In [1]:
```
import sagemaker
from sagemaker import get_execution_role
```

```
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemake
r/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.c
onfig/sagemaker/config.yaml
```

In [2]:
```
import pandas as pd

df = pd.read_csv("final_airbnb_ml.csv")
df.head()
```

Out[2]:

	306	4	4.8	531	FALSE	FALSE.1	FALSE.2	FALSE.3	FALSE.4	FALSE.5	...	FALSE.655	FALSE.65
0	485	4	4.7	146	False	False	False	False	True	False	...	False	Fal
1	119	4	4.9	515	False	False	False	False	True	False	...	False	Fal
2	192	5	4.9	88	False	False	False	False	False	False	...	False	Fal
3	232	1	4.9	222	False	False	False	False	False	False	...	False	Fal
4	261	1	4.8	555	False	False	False	False	False	False	...	False	Fal

5 rows × 671 columns

In [3]:
```
print(df.shape)
print(df.columns.tolist())
```

```
(952, 671)
['306', '4', '4.8', '531', 'FALSE', 'FALSE.1', 'FALSE.2', 'FALSE.3', 'FALSE.4', 'F
ALSE.5', 'FALSE.6', 'FALSE.7', 'FALSE.8', 'TRUE', 'FALSE.9', 'FALSE.10', 'FALSE.1
1', 'FALSE.12', 'FALSE.13', 'FALSE.14', 'FALSE.15', 'FALSE.16', 'FALSE.17', 'FALS
E.18', 'FALSE.19', 'FALSE.20', 'FALSE.21', 'FALSE.22', 'FALSE.23', 'FALSE.24', 'FA
LSE.25', 'FALSE.26', 'FALSE.27', 'FALSE.28', 'FALSE.29', 'FALSE.30', 'FALSE.31',
'FALSE.32', 'FALSE.33', 'FALSE.34', 'FALSE.35', 'FALSE.36', 'FALSE.37', 'FALSE.3
8', 'FALSE.39', 'FALSE.40', 'FALSE.41', 'FALSE.42', 'FALSE.43', 'FALSE.44', 'FALS
E.45', 'FALSE.46', 'FALSE.47', 'FALSE.48', 'FALSE.49', 'FALSE.50', 'FALSE.51', 'FA
LSE.52', 'FALSE.53', 'FALSE.54', 'FALSE.55', 'FALSE.56', 'FALSE.57', 'FALSE.58',
'FALSE.59', 'FALSE.60', 'FALSE.61', 'FALSE.62', 'FALSE.63', 'FALSE.64', 'FALSE.6
5', 'FALSE.66', 'FALSE.67', 'FALSE.68', 'FALSE.69', 'FALSE.70', 'FALSE.71', 'FALS
E.72', 'FALSE.73', 'FALSE.74', 'FALSE.75', 'FALSE.76', 'FALSE.77', 'FALSE.78', 'FA
LSE.79', 'FALSE.80', 'FALSE.81', 'FALSE.82', 'FALSE.83', 'FALSE.84', 'FALSE.85',
'FALSE.86', 'FALSE.87', 'FALSE.88', 'FALSE.89', 'FALSE.90', 'FALSE.91', 'FALSE.9
2', 'FALSE.93', 'FALSE.94', 'FALSE.95', 'FALSE.96', 'FALSE.97', 'FALSE.98', 'FALS
E.99', 'FALSE.100', 'FALSE.101', 'FALSE.102', 'FALSE.103', 'FALSE.104', 'FALSE.10
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'FALSE.112', 'FALSE.113', 'FALSE.114', 'FALSE.115', 'FALSE.116', 'FALSE.117', 'FAL
SE.118', 'FALSE.119', 'FALSE.120', 'FALSE.121', 'FALSE.122', 'FALSE.123', 'FALSE.1
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SE.270', 'FALSE.271', 'FALSE.272', 'FALSE.273', 'FALSE.274', 'FALSE.275', 'FALSE.2
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```

```
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SE.649', 'FALSE.650', 'FALSE.651', 'FALSE.652', 'FALSE.653', 'FALSE.654', 'FALSE.6
55', 'FALSE.656', 'FALSE.657', 'FALSE.658', 'FALSE.659', 'FALSE.660', 'FALSE.661',
'FALSE.662', 'FALSE.663', 'FALSE.664']
```

In [4]:

```
df = df.replace({True: 1, False: 0})
df.head()
```

```
/tmp/ipykernel_9467/1757821305.py:1: FutureWarning: Downcasting behavior in `repla
ce` is deprecated and will be removed in a future version. To retain the old behav
ior, explicitly call `result.infer_objects(copy=False)`. To opt-in to the future b
ehavior, set `pd.set_option('future.no_silent_downcasting', True)`
  df = df.replace({True: 1, False: 0})
```

Out[4]:

	306	4	4.8	531	FALSE	FALSE.1	FALSE.2	FALSE.3	FALSE.4	FALSE.5	...	FALSE.655	FALSE.65
0	485	4	4.7	146	0	0	0	0	1	0	...	0	
1	119	4	4.9	515	0	0	0	0	1	0	...	0	
2	192	5	4.9	88	0	0	0	0	0	0	...	0	
3	232	1	4.9	222	0	0	0	0	0	0	...	0	
4	261	1	4.8	555	0	0	0	0	0	0	...	0	

5 rows × 671 columns

In [5]:

```
cols = ['price(in_dollar)'] + [col for col in df.columns if col != 'price(in_dollar
df = df[cols]

df.to_csv("sagemaker_train.csv", index=False, header=False)
```

```
Traceback (most recent call last)
in <module>:2

  1 cols = ['price(in_dollar)'] + [col for col in df.columns if col != 'pric
❯ 2 df = df[cols]
  3
  4 df.to_csv("sagemaker_train.csv", index=False, header=False)
  5

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/pandas/co
__getitem__

   4116 │   │   else:
   4117 │   │   │   if is_iterator(key):
   4118 │   │   │   │   key = list(key)
❯  4119 │   │   │   indexer = self.columns._get_indexer_strict(key, "columns
   4120 │   │
   4121 │   │   # take() does not accept boolean indexers
   4122 │   │   if getattr(indexer, "dtype", None) == bool:

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/pandas/co
212 in _get_indexer_strict

  6209 │   │   else:
  6210 │   │   │   keyarr, indexer, new_indexer = self._reindex_non_unique(k
  6211 │   │
❯ 6212 │   │   self._raise_if_missing(keyarr, indexer, axis_name)
  6213 │   │
  6214 │   │   keyarr = self.take(indexer)
  6215 │   │   if isinstance(key, Index):

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/pandas/co
264 in _raise_if_missing

  6261 │   │   │   │   raise KeyError(f"None of [{key}] are in the [{axis_na
  6262 │   │   │
  6263 │   │   │   not_found = list(ensure_index(key)[missing_mask.nonzero()
❯ 6264 │   │   │   raise KeyError(f"{not_found} not in index")
  6265 │
  6266 │   @overload
  6267 │   def _get_indexer_non_comparable(
```

KeyError: "['price(in_dollar)'] not in index"

In [6]:
```
with open("sagemaker_train.csv", "r") as f:
    for i in range(3):
        print(f.readline())
```

```
Traceback (most recent call last)
in <module>:1

❯ 1 with open("sagemaker_train.csv", "r") as f:
  2 |   for i in range(3):
  3 |   |   print(f.readline())
  4

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/IPython/c
.py:344 in _modified_open

  341 |   |   |   "you can use builtins' open."
  342 |   |   )
  343 |
❯ 344 |   return io_open(file, *args, **kwargs)
  345
  346
  347 class InteractiveShell(SingletonConfigurable):
```

```
FileNotFoundError: [Errno 2] No such file or directory: 'sagemaker_train.csv'
```

In [8]:
```
print('price(in_dollar)' in df.columns)
```

```
False
```

In [9]:
```
df.to_csv("sagemaker_train.csv", index=False, header=False)
```

In [10]:
```
import os
print(os.listdir())
```

```
['final_airbnb_ml.csv', 'lost+found', '.ipynb_checkpoints', '.virtual_documents',
'airbnb.csv', 'sagemaker_train.csv', 'Untitled1.ipynb', '.sparkmagic', '.Trash-100
0']
```

In [11]:
```
import sagemaker
from sagemaker import get_execution_role

session = sagemaker.Session()
bucket = session.default_bucket()
role = get_execution_role()

print("Bucket:", bucket)
print("Role:", role)
```

```
Bucket: sagemaker-ca-central-1-263245924849
Role: arn:aws:iam::263245924849:role/service-role/AmazonSageMaker-ExecutionRole-20
260323T193815
```

In [12]:
```
s3_path = session.upload_data(
    path="sagemaker_train.csv",
    bucket=bucket,
    key_prefix="airbnb-price-project"
)

print("S3 Path:", s3_path)
```

```
S3 Path: s3://sagemaker-ca-central-1-263245924849/airbnb-price-project/sagemaker_t
rain.csv
```

In [13]:
```
train_df = pd.read_csv("sagemaker_train.csv", header=None)
train_df.head()
```

Out[13]:

	0	1	2	3	4	5	6	7	8	9	...	661	662	663	664	665	666	667	668	669	670
0	485	4	4.7	146	0	0	0	0	1	0	...	0	0	0	0	0	0	0	0	0	0
1	119	4	4.9	515	0	0	0	0	1	0	...	0	0	0	0	0	0	0	0	0	0
2	192	5	4.9	88	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
3	232	1	4.9	222	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
4	261	1	4.8	555	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0

5 rows × 671 columns

In [14]:
```
X = train_df.iloc[:, 1:]
y = train_df.iloc[:, 0]
```

In [15]:
```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

In [16]:
```
from sklearn.ensemble import RandomForestRegressor

rf_model = RandomForestRegressor(random_state=42)
rf_model.fit(X_train, y_train)
```

Out[16]:
```
▾ RandomForestRegressor
▸ Parameters
```

In [17]:
```
from sklearn.metrics import mean_absolute_error, r2_score

rf_pred = rf_model.predict(X_test)

print("MAE:", mean_absolute_error(y_test, rf_pred))
print("R2 Score:", r2_score(y_test, rf_pred))
```

```
MAE: 74.70068062827224
R2 Score: 0.2725460882792329
```

In [18]:
```
import joblib

joblib.dump(rf_model, "airbnb_price_model.pkl")
print("Model saved successfully.")
```

```
Model saved successfully.
```

In [19]:
```
import os
print(os.listdir())
```

```
['final_airbnb_ml.csv', 'lost+found', '.ipynb_checkpoints', '.virtual_documents',
'airbnb.csv', 'airbnb_price_model.pkl', 'sagemaker_train.csv', 'Untitled1.ipynb',
'.sparkmagic', '.Trash-1000']
```

In [23]:
```
!pip install seaborn
```

```
Collecting seaborn
  Downloading seaborn-0.13.2-py3-none-any.whl.metadata (5.4 kB)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in /home/ec2-user/anaconda3/en
vs/python3/lib/python3.12/site-packages (from seaborn) (1.26.4)
Requirement already satisfied: pandas>=1.2 in /home/ec2-user/anaconda3/envs/python
3/lib/python3.12/site-packages (from seaborn) (2.3.3)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in /home/ec2-user/anaconda
3/envs/python3/lib/python3.12/site-packages (from seaborn) (3.10.8)
Requirement already satisfied: contourpy>=1.0.1 in /home/ec2-user/anaconda3/envs/p
ython3/lib/python3.12/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.
3)
Requirement already satisfied: cycler>=0.10 in /home/ec2-user/anaconda3/envs/pytho
n3/lib/python3.12/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /home/ec2-user/anaconda3/envs/
python3/lib/python3.12/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.6
1.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /home/ec2-user/anaconda3/envs/
python3/lib/python3.12/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.
9)
Requirement already satisfied: packaging>=20.0 in /home/ec2-user/anaconda3/envs/py
thon3/lib/python3.12/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (24.2)
Requirement already satisfied: pillow>=8 in /home/ec2-user/anaconda3/envs/python3/
lib/python3.12/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.1.0)
Requirement already satisfied: pyparsing>=3 in /home/ec2-user/anaconda3/envs/pytho
n3/lib/python3.12/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /home/ec2-user/anaconda3/en
vs/python3/lib/python3.12/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /home/ec2-user/anaconda3/envs/pytho
n3/lib/python3.12/site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /home/ec2-user/anaconda3/envs/pyt
hon3/lib/python3.12/site-packages (from pandas>=1.2->seaborn) (2025.3)
Requirement already satisfied: six>=1.5 in /home/ec2-user/anaconda3/envs/python3/l
ib/python3.12/site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->s
eaborn) (1.17.0)
Downloading seaborn-0.13.2-py3-none-any.whl (294 kB)
Installing collected packages: seaborn
Successfully installed seaborn-0.13.2
```

In [24]:
```
import matplotlib.pyplot as plt
import seaborn as sns
```

In [25]:
```
plt.figure(figsize=(8,5))
sns.histplot(df['price(in_dollar)'], bins=30, kde=True)
plt.title("Distribution of Airbnb Prices")
plt.xlabel("Price (in Dollar)")
```

```
plt.ylabel("Count")
plt.show()
```

```
Traceback (most recent call last)
/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/pandas/co
812 in get_loc

  3809 │   │   """
  3810 │   │   casted_key = self._maybe_cast_indexer(key)
  3811 │   │   try:
❯ 3812 │   │   │   return self._engine.get_loc(casted_key)
  3813 │   │   except KeyError as err:
  3814 │   │   │   if isinstance(casted_key, slice) or (
  3815 │   │   │   │   isinstance(casted_key, abc.Iterable)

in pandas._libs.index.IndexEngine.get_loc:167

in pandas._libs.index.IndexEngine.get_loc:196

in pandas._libs.hashtable.PyObjectHashTable.get_item:7088

in pandas._libs.hashtable.PyObjectHashTable.get_item:7096
```

KeyError: 'price(in_dollar)'

The above exception was the direct cause of the following exception:

```
Traceback (most recent call last)
in <module>:2

  1 plt.figure(figsize=(8,5))
❯ 2 sns.histplot(df['price(in_dollar)'], bins=30, kde=True)
  3 plt.title("Distribution of Airbnb Prices")
  4 plt.xlabel("Price (in Dollar)")
  5 plt.ylabel("Count")

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/pandas/co
__getitem__

   4110 │   │   if is_single_key:
   4111 │   │   │   if self.columns.nlevels > 1:
   4112 │   │   │   │   return self._getitem_multilevel(key)
❯  4113 │   │   │   indexer = self.columns.get_loc(key)
   4114 │   │   │   if is_integer(indexer):
   4115 │   │   │   │   indexer = [indexer]
   4116 │   │   else:

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/pandas/co
819 in get_loc

  3816 │   │   │   │   and any(isinstance(x, slice) for x in casted_key)
  3817 │   │   │   ):
  3818 │   │   │   │   raise InvalidIndexError(key)
❯ 3819 │   │   │   raise KeyError(key) from err
  3820 │   │   except TypeError:
  3821 │   │   │   # If we have a listlike key, _check_indexing_error will r
  3822 │   │   │   #  InvalidIndexError. Otherwise we fall through and re-ra
```

KeyError: 'price(in_dollar)'

In [26]:
```
plt.figure(figsize=(8,5))
sns.scatterplot(x=df['beds'], y=df['price(in_dollar)'])
plt.title("Beds vs Airbnb Price")
plt.xlabel("Number of Beds")
plt.ylabel("Price (in Dollar)")
plt.show()
```

```
Traceback (most recent call last)
/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/pandas/co
812 in get_loc

  3809 │   │   """
  3810 │   │   casted_key = self._maybe_cast_indexer(key)
  3811 │   │   try:
❯ 3812 │   │   │   return self._engine.get_loc(casted_key)
  3813 │   │   except KeyError as err:
  3814 │   │   │   if isinstance(casted_key, slice) or (
  3815 │   │   │   │   isinstance(casted_key, abc.Iterable)

in pandas._libs.index.IndexEngine.get_loc:167

in pandas._libs.index.IndexEngine.get_loc:196

in pandas._libs.hashtable.PyObjectHashTable.get_item:7088

in pandas._libs.hashtable.PyObjectHashTable.get_item:7096
```

KeyError: 'beds'

The above exception was the direct cause of the following exception:

```
Traceback (most recent call last)
in <module>:2

  1 plt.figure(figsize=(8,5))
❯ 2 sns.scatterplot(x=df['beds'], y=df['price(in_dollar)'])
  3 plt.title("Beds vs Airbnb Price")
  4 plt.xlabel("Number of Beds")
  5 plt.ylabel("Price (in Dollar)")

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/pandas/co
__getitem__

   4110 │   │   if is_single_key:
   4111 │   │   │   if self.columns.nlevels > 1:
   4112 │   │   │   │   return self._getitem_multilevel(key)
❯  4113 │   │   │   indexer = self.columns.get_loc(key)
   4114 │   │   │   if is_integer(indexer):
   4115 │   │   │   │   indexer = [indexer]
   4116 │   │   else:

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/pandas/co
819 in get_loc

  3816 │   │   │   │   and any(isinstance(x, slice) for x in casted_key)
  3817 │   │   │   ):
  3818 │   │   │   │   raise InvalidIndexError(key)
❯ 3819 │   │   │   raise KeyError(key) from err
  3820 │   │   except TypeError:
  3821 │   │   │   # If we have a listlike key, _check_indexing_error will r
  3822 │   │   │   #  InvalidIndexError. Otherwise we fall through and re-ra
```

KeyError: 'beds'

In [27]:
```
plt.figure(figsize=(8,5))
sns.scatterplot(x=df['rating'], y=df['price(in_dollar)'])
plt.title("Rating vs Airbnb Price")
plt.xlabel("Rating")
plt.ylabel("Price (in Dollar)")
plt.show()
```

```
Traceback (most recent call last)
/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/pandas/co
812 in get_loc

  3809 │   │   """
  3810 │   │   casted_key = self._maybe_cast_indexer(key)
  3811 │   │   try:
❯ 3812 │   │   │   return self._engine.get_loc(casted_key)
  3813 │   │   except KeyError as err:
  3814 │   │   │   if isinstance(casted_key, slice) or (
  3815 │   │   │   │   isinstance(casted_key, abc.Iterable)

in pandas._libs.index.IndexEngine.get_loc:167

in pandas._libs.index.IndexEngine.get_loc:196

in pandas._libs.hashtable.PyObjectHashTable.get_item:7088

in pandas._libs.hashtable.PyObjectHashTable.get_item:7096
```

KeyError: 'rating'

The above exception was the direct cause of the following exception:

```
Traceback (most recent call last)
in <module>:2

  1 plt.figure(figsize=(8,5))
❯ 2 sns.scatterplot(x=df['rating'], y=df['price(in_dollar)'])
  3 plt.title("Rating vs Airbnb Price")
  4 plt.xlabel("Rating")
  5 plt.ylabel("Price (in Dollar)")

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/pandas/co
__getitem__

   4110 │   │   if is_single_key:
   4111 │   │   │   if self.columns.nlevels > 1:
   4112 │   │   │   │   return self._getitem_multilevel(key)
❯  4113 │   │   │   indexer = self.columns.get_loc(key)
   4114 │   │   │   if is_integer(indexer):
   4115 │   │   │   │   indexer = [indexer]
   4116 │   │   else:

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/pandas/co
819 in get_loc

  3816 │   │   │   │   and any(isinstance(x, slice) for x in casted_key)
  3817 │   │   │   ):
  3818 │   │   │   │   raise InvalidIndexError(key)
❯ 3819 │   │   │   raise KeyError(key) from err
  3820 │   │   except TypeError:
  3821 │   │   │   # If we have a listlike key, _check_indexing_error will r
  3822 │   │   │   #  InvalidIndexError. Otherwise we fall through and re-ra
```

KeyError: 'rating'

In [28]:
```
[col for col in df.columns if 'rating' in col.lower()]
```

Out[28]:
```
[]
```

In [29]:
```
plt.figure(figsize=(8,5))
sns.scatterplot(x=df['num_reviews'], y=df['price(in_dollar)'])
plt.title("Number of Reviews vs Airbnb Price")
plt.xlabel("Number of Reviews")
plt.ylabel("Price (in Dollar)")
plt.show()
```

```
Traceback (most recent call last)
/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/pandas/co
812 in get_loc

  3809 |   |   """
  3810 |   |   casted_key = self._maybe_cast_indexer(key)
  3811 |   |   try:
❯ 3812 |   |   |   return self._engine.get_loc(casted_key)
  3813 |   |   except KeyError as err:
  3814 |   |   |   if isinstance(casted_key, slice) or (
  3815 |   |   |   |   isinstance(casted_key, abc.Iterable)

in pandas._libs.index.IndexEngine.get_loc:167

in pandas._libs.index.IndexEngine.get_loc:196

in pandas._libs.hashtable.PyObjectHashTable.get_item:7088

in pandas._libs.hashtable.PyObjectHashTable.get_item:7096
```

KeyError: 'num_reviews'

The above exception was the direct cause of the following exception:

```
Traceback (most recent call last)
in <module>:2

  1 plt.figure(figsize=(8,5))
❯ 2 sns.scatterplot(x=df['num_reviews'], y=df['price(in_dollar)'])
  3 plt.title("Number of Reviews vs Airbnb Price")
  4 plt.xlabel("Number of Reviews")
  5 plt.ylabel("Price (in Dollar)")

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/pandas/co
__getitem__

   4110 |   |   if is_single_key:
   4111 |   |   |   if self.columns.nlevels > 1:
   4112 |   |   |   |   return self._getitem_multilevel(key)
❯  4113 |   |   |   indexer = self.columns.get_loc(key)
   4114 |   |   |   if is_integer(indexer):
   4115 |   |   |   |   indexer = [indexer]
   4116 |   |   else:

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/pandas/co
819 in get_loc

  3816 |   |   |   |   and any(isinstance(x, slice) for x in casted_key)
  3817 |   |   |   ):
  3818 |   |   |   |   raise InvalidIndexError(key)
❯ 3819 |   |   |   raise KeyError(key) from err
  3820 |   |   except TypeError:
  3821 |   |   |   # If we have a listlike key, _check_indexing_error will r
  3822 |   |   |   #  InvalidIndexError. Otherwise we fall through and re-ra
```

KeyError: 'num_reviews'

In [30]:
```
[col for col in df.columns if 'review' in col.lower()]
```

Out[30]:
```
[]
```

In [31]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("final_airbnb_ml.csv")
df = df.replace({True: 1, False: 0})

print(df.shape)
df.head()
```

```
(952, 671)
/tmp/ipykernel_9467/3740061375.py:6: FutureWarning: Downcasting behavior in `repla
ce` is deprecated and will be removed in a future version. To retain the old behav
ior, explicitly call `result.infer_objects(copy=False)`. To opt-in to the future b
ehavior, set `pd.set_option('future.no_silent_downcasting', True)`
  df = df.replace({True: 1, False: 0})
```

Out[31]:

	306	4	4.8	531	FALSE	FALSE.1	FALSE.2	FALSE.3	FALSE.4	FALSE.5	...	FALSE.655	FALSE.65
0	485	4	4.7	146	0	0	0	0	1	0	...	0	
1	119	4	4.9	515	0	0	0	0	1	0	...	0	
2	192	5	4.9	88	0	0	0	0	0	0	...	0	
3	232	1	4.9	222	0	0	0	0	0	0	...	0	
4	261	1	4.8	555	0	0	0	0	0	0	...	0	

5 rows × 671 columns

In [32]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("final_airbnb_ml.csv")
print(df.shape)
print(df.columns.tolist()[:10])
df.head()
```

```
(952, 671)
['306', '4', '4.8', '531', 'FALSE', 'FALSE.1', 'FALSE.2', 'FALSE.3', 'FALSE.4', 'F
ALSE.5']
```

Out[32]:

	306	4	4.8	531	FALSE	FALSE.1	FALSE.2	FALSE.3	FALSE.4	FALSE.5	...	FALSE.655	FALSE.65
0	485	4	4.7	146	False	False	False	False	True	False	...	False	Fal
1	119	4	4.9	515	False	False	False	False	True	False	...	False	Fal
2	192	5	4.9	88	False	False	False	False	False	False	...	False	Fal
3	232	1	4.9	222	False	False	False	False	False	False	...	False	Fal
4	261	1	4.8	555	False	False	False	False	False	False	...	False	Fal

5 rows × 671 columns

In [34]:

```
df = pd.read_csv("final_airbnb_ml.csv", header=None)

# rename first important columns
new_cols = ['price(in_dollar)', 'beds', 'rating', 'num_reviews'] + [f'feature_{i}'
df.columns = new_cols

print(df.shape)
print(df.columns[:10])
df.head()
```

```
(953, 671)
Index(['price(in_dollar)', 'beds', 'rating', 'num_reviews', 'feature_4',
       'feature_5', 'feature_6', 'feature_7', 'feature_8', 'feature_9'],
      dtype='object')
```

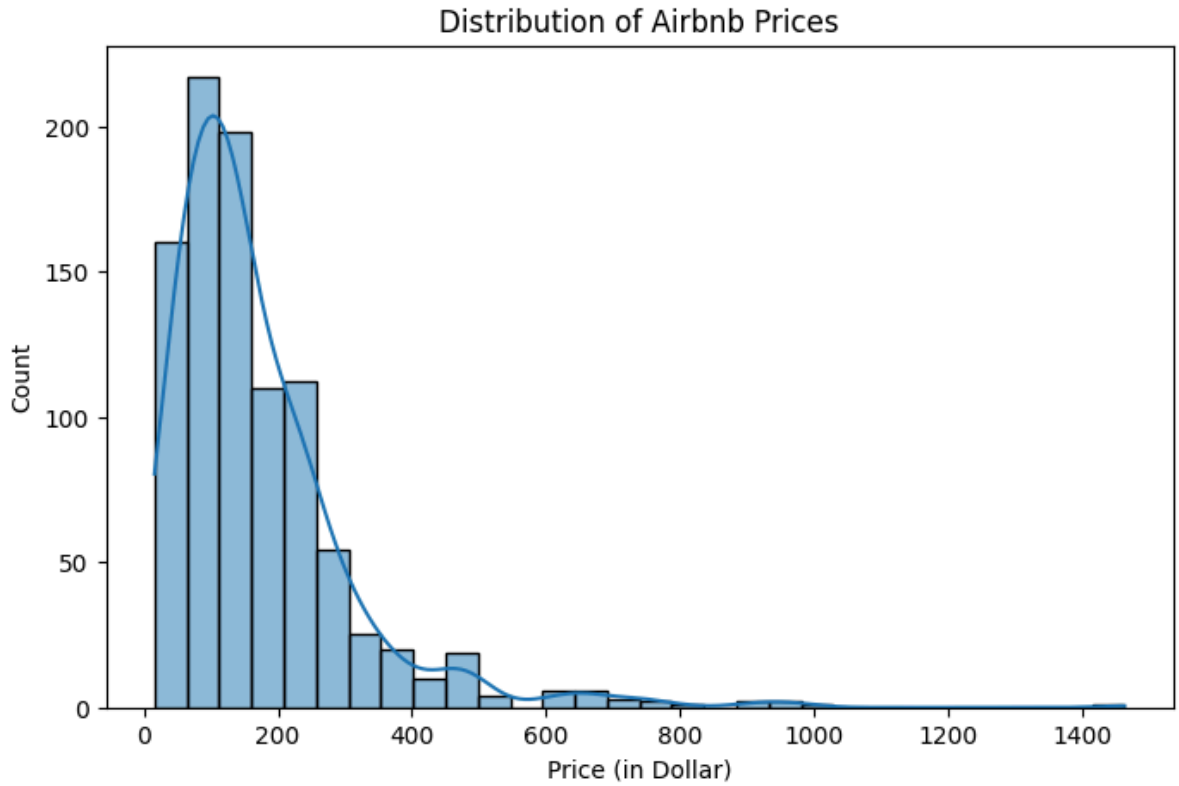
Out[34]:

	price(in_dollar)	beds	rating	num_reviews	feature_4	feature_5	feature_6	feature_7	feature_
0	306	4	4.8	531	False	False	False	False	Fals
1	485	4	4.7	146	False	False	False	False	Tru
2	119	4	4.9	515	False	False	False	False	Tru
3	192	5	4.9	88	False	False	False	False	Fals
4	232	1	4.9	222	False	False	False	False	Fals

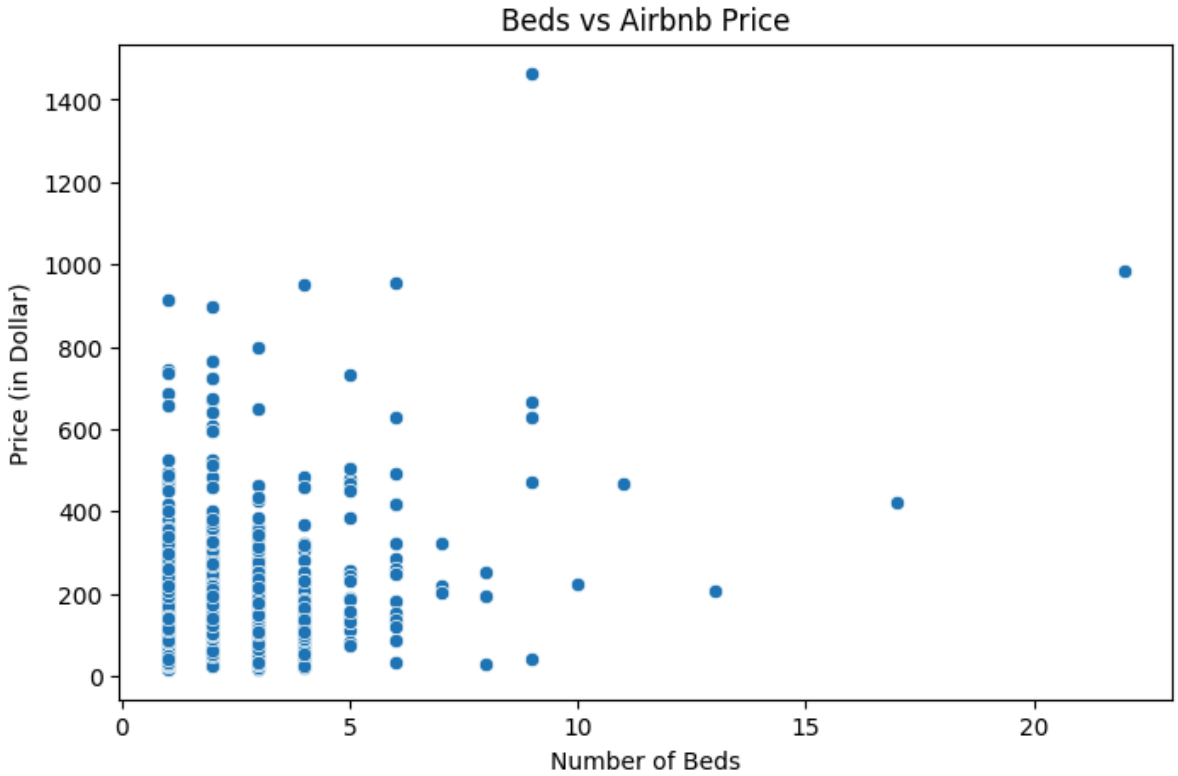
5 rows × 671 columns

In [35]:

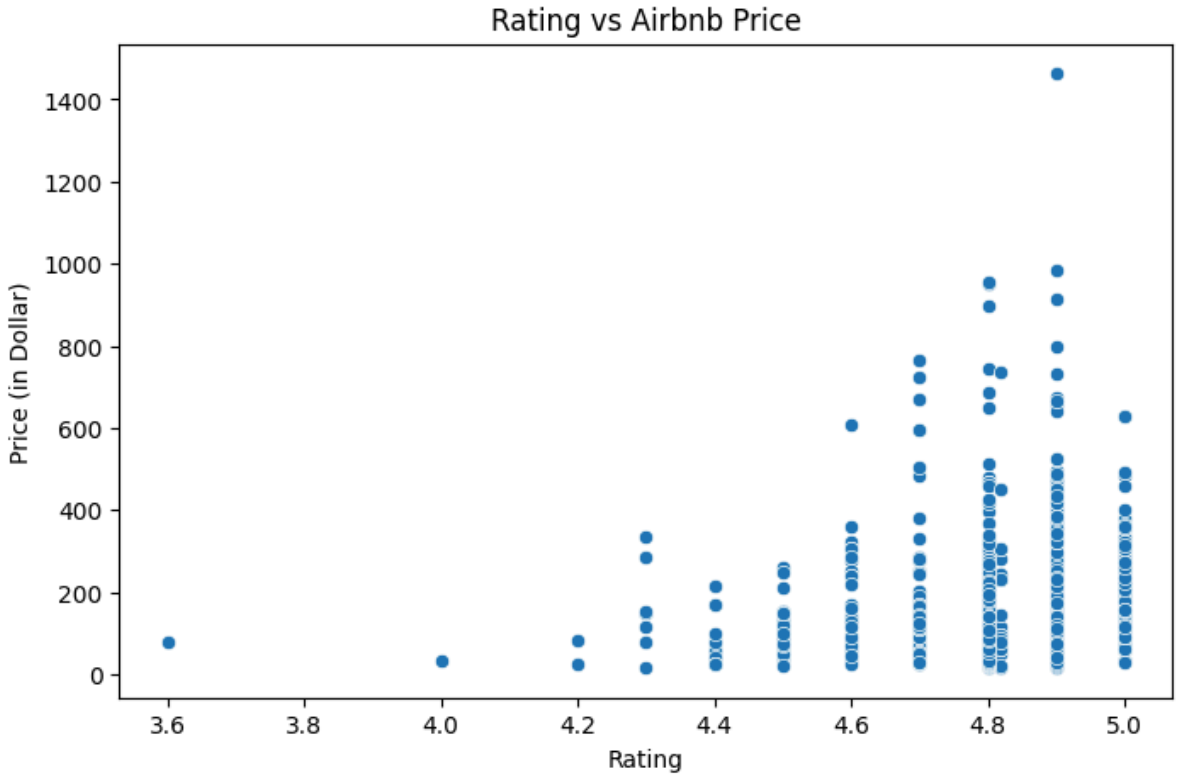
```
plt.figure(figsize=(8,5))
sns.histplot(df['price(in_dollar)'], bins=30, kde=True)
plt.title("Distribution of Airbnb Prices")
plt.xlabel("Price (in Dollar)")
plt.ylabel("Count")
plt.show()
```



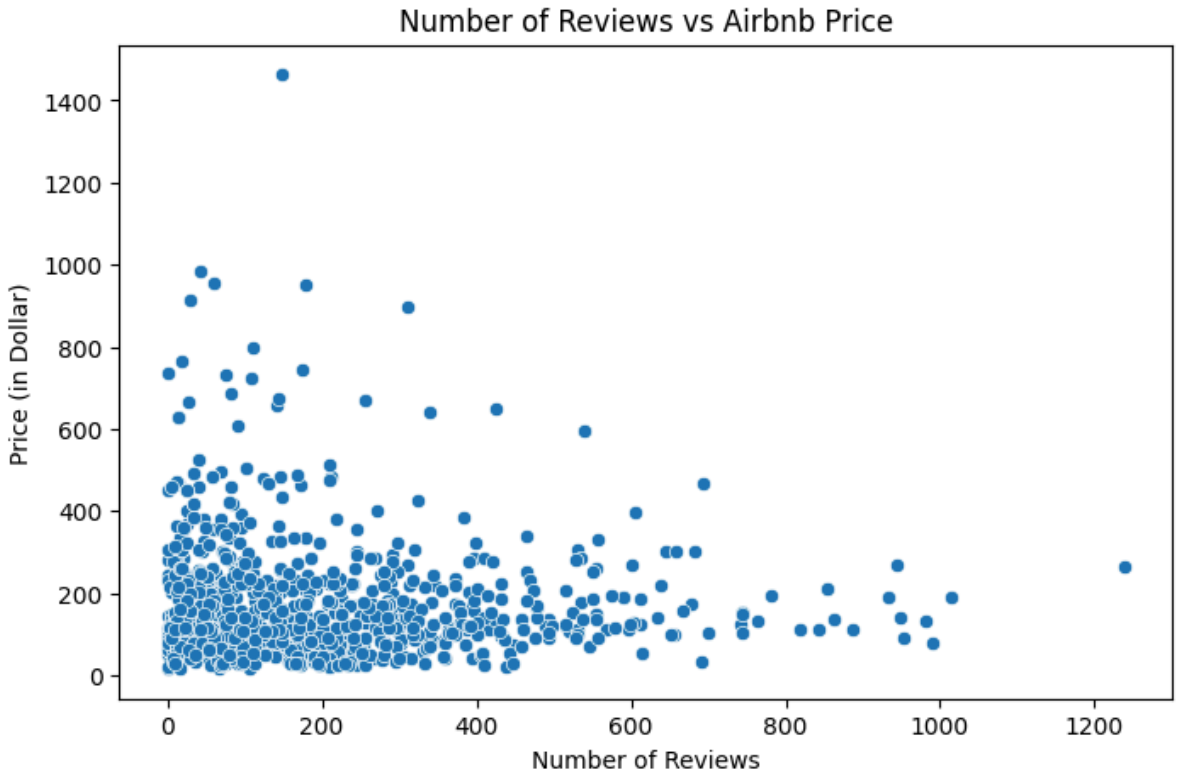
In [36]:
```
plt.figure(figsize=(8,5))
sns.scatterplot(x=df['beds'], y=df['price(in_dollar)'])
plt.title("Beds vs Airbnb Price")
plt.xlabel("Number of Beds")
plt.ylabel("Price (in Dollar)")
plt.show()
```



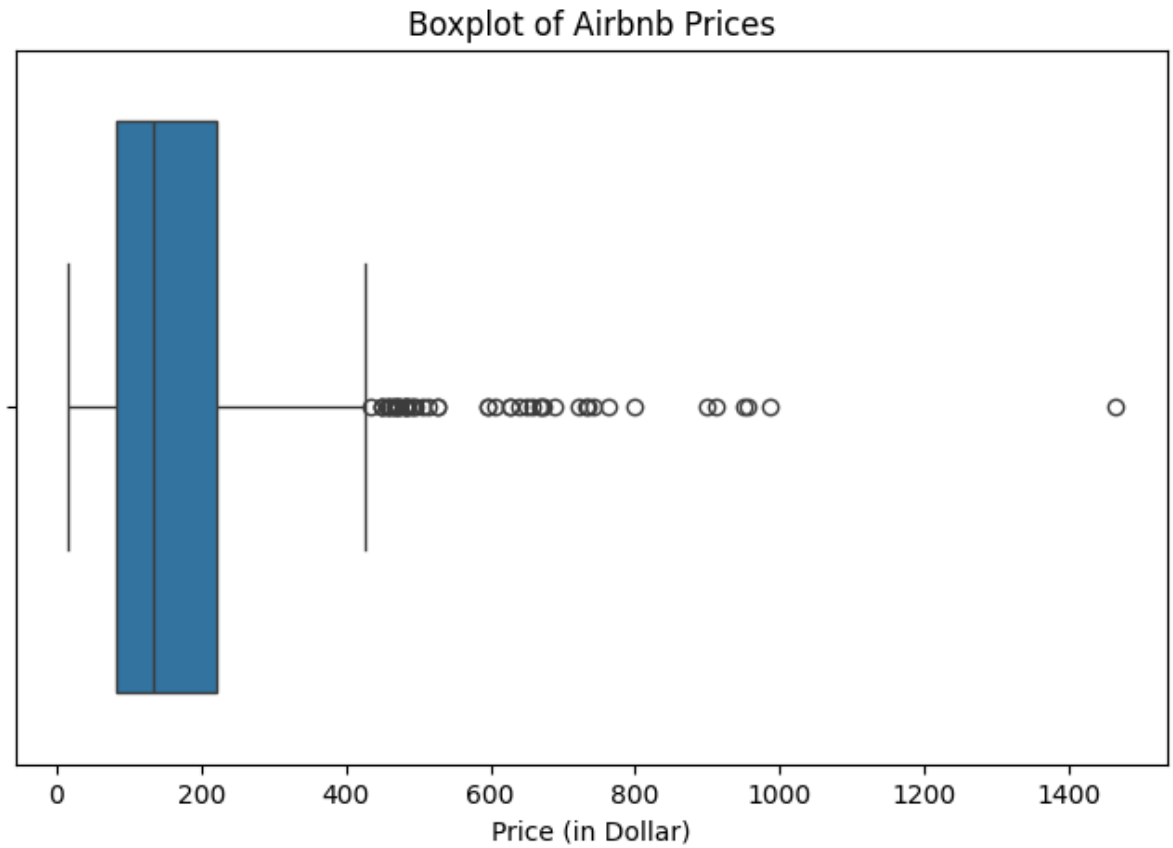
In [37]:
```
plt.figure(figsize=(8,5))
sns.scatterplot(x=df['rating'], y=df['price(in_dollar)'])
plt.title("Rating vs Airbnb Price")
plt.xlabel("Rating")
plt.ylabel("Price (in Dollar)")
plt.show()
```



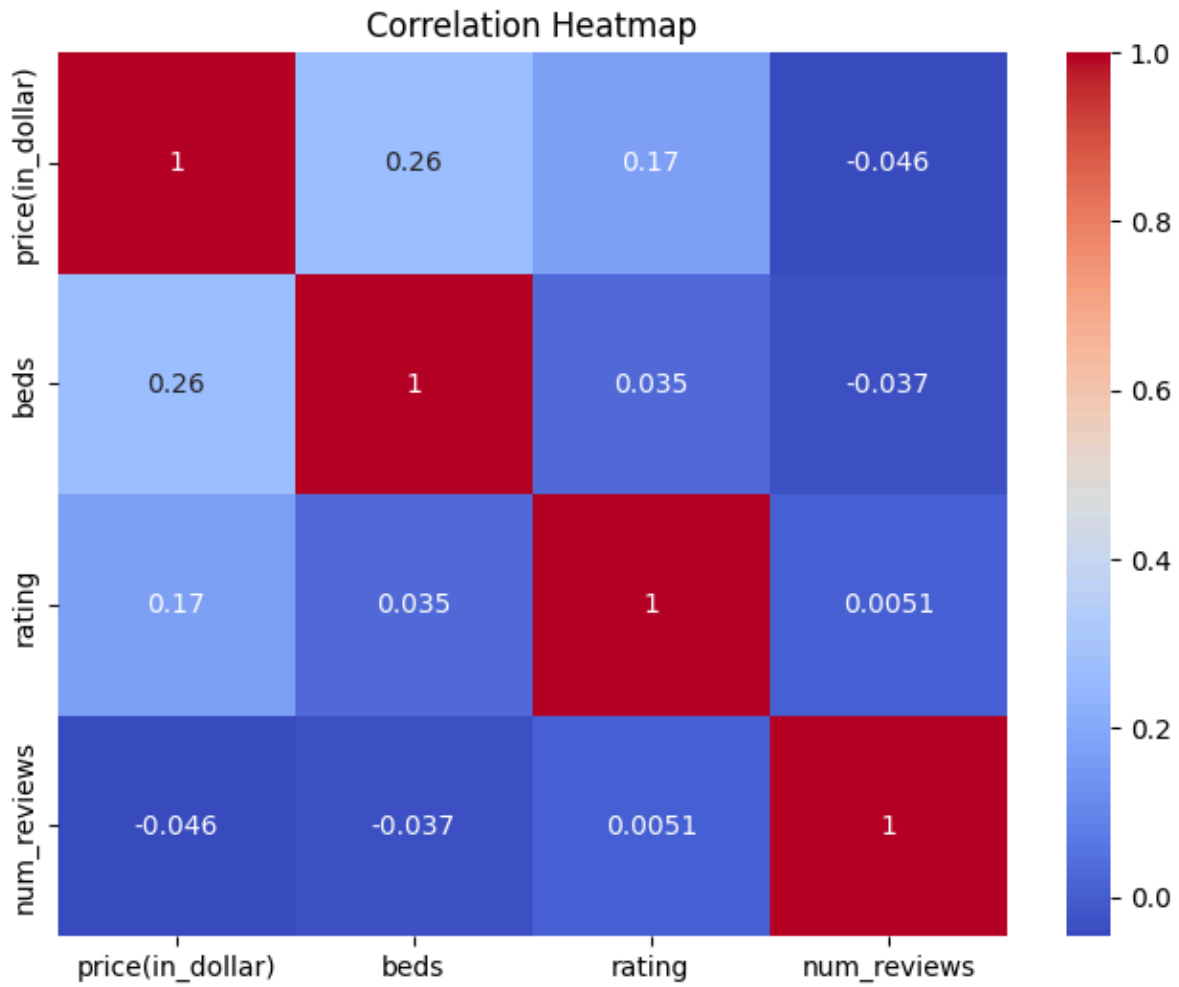
In [38]:
```
plt.figure(figsize=(8,5))
sns.scatterplot(x=df['num_reviews'], y=df['price(in_dollar)'])
plt.title("Number of Reviews vs Airbnb Price")
plt.xlabel("Number of Reviews")
plt.ylabel("Price (in Dollar)")
plt.show()
```



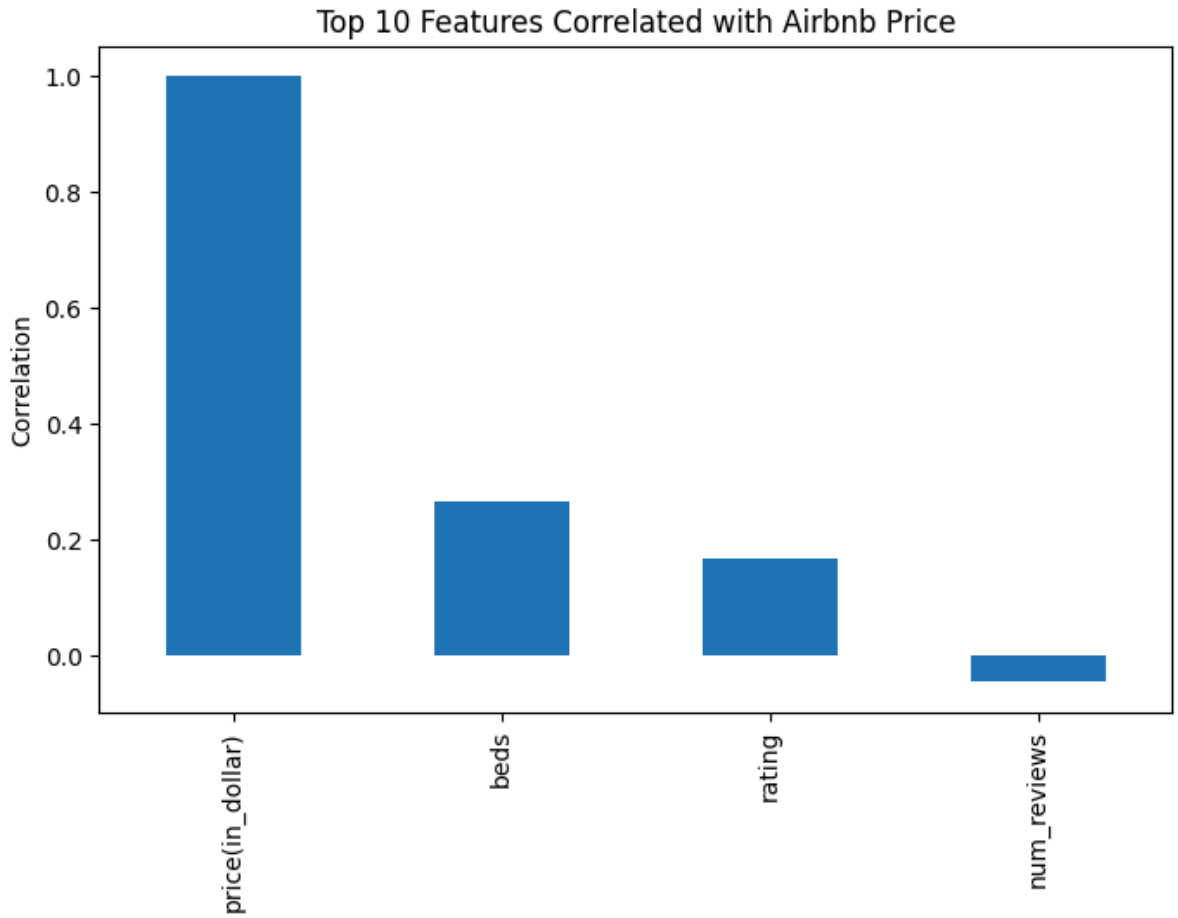
In [39]:
```
plt.figure(figsize=(8,5))
sns.boxplot(x=df['price(in_dollar)'])
plt.title("Boxplot of Airbnb Prices")
plt.xlabel("Price (in Dollar)")
plt.show()
```



In [40]:
```
numeric_df = df.select_dtypes(include=['int64', 'float64'])
corr = numeric_df[['price(in_dollar)', 'beds', 'rating', 'num_reviews']].corr()

plt.figure(figsize=(8,6))
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



In [41]:
```
top_corr = numeric_df.corr()['price(in_dollar)'].sort_values(ascending=False).head(

plt.figure(figsize=(8,5))
top_corr.plot(kind='bar')
plt.title("Top 10 Features Correlated with Airbnb Price")
plt.ylabel("Correlation")
plt.show()
```



In [42]:
```
import os

os.makedirs("model", exist_ok=True)
```

In [43]:
```
import joblib

joblib.dump(rf_model, "model/model.joblib")
print("Model saved inside model folder.")
```

```
Model saved inside model folder.
```

In [44]:
```
import tarfile

with tarfile.open("model.tar.gz", "w:gz") as tar:
    tar.add("model", arcname=".")

print("model.tar.gz created successfully.")
```

```
model.tar.gz created successfully.
```

In [45]:
```
print(os.listdir())
print(os.listdir("model"))
```

```
['final_airbnb_ml.csv', 'lost+found', '.ipynb_checkpoints', '.virtual_documents',
'airbnb.csv', 'model', 'airbnb_price_model.pkl', 'model.tar.gz', 'sagemaker_train.
csv', 'Untitled1.ipynb', '.sparkmagic', '.Trash-1000']
['model.joblib']
```

In [46]:
```
model_s3_path = session.upload_data(
    path="model.tar.gz",
    bucket=bucket,
    key_prefix="airbnb-price-model"
)

print("Model S3 Path:", model_s3_path)
```

```
Model S3 Path: s3://sagemaker-ca-central-1-263245924849/airbnb-price-model/model.t
ar.gz
```

In [47]:
```
model_s3_path = session.upload_data(
    path="model.tar.gz",
    bucket=bucket,
    key_prefix="airbnb-price-model"
)

print("Model S3 Path:", model_s3_path)
```

```
Model S3 Path: s3://sagemaker-ca-central-1-263245924849/airbnb-price-model/model.t
ar.gz
```

In [48]:
```
%%writefile inference.py

import joblib
import numpy as np

def model_fn(model_dir):
    model = joblib.load(f"{model_dir}/model.joblib")
    return model

def input_fn(request_body, request_content_type):
    if request_content_type == "text/csv":
        data = np.array([float(x) for x in request_body.split(",")])
        return data.reshape(1, -1)
    else:
        raise ValueError("Unsupported content type")

def predict_fn(input_data, model):
    prediction = model.predict(input_data)
    return prediction

def output_fn(prediction, content_type):
    return str(prediction[0])
```

```
Writing inference.py
```

In [49]:
```
from sagemaker.sklearn.model import SKLearnModel

sk_model = SKLearnModel(
    model_data=model_s3_path,
    role=role,
    entry_point="inference.py",
    framework_version="1.2-1"
)
```

In [51]:
```
from sagemaker import Session

try:
    predictor.delete_endpoint()
    print("Broken endpoint deleted.")
except:
    print("No active endpoint to delete.")
```

```
No active endpoint to delete.
```

In [52]:
```
import joblib

joblib.dump(rf_model, "model.joblib")
print("Saved model.joblib")
```

```
Saved model.joblib
```

In [53]:
```
import tarfile

with tarfile.open("model.tar.gz", "w:gz") as tar:
    tar.add("model.joblib")

print("New model.tar.gz created correctly.")
```

```
New model.tar.gz created correctly.
```

In [54]:
```
model_s3_path = session.upload_data(
    path="model.tar.gz",
    bucket=bucket,
    key_prefix="airbnb-price-model-fixed"
)

print("Fixed Model S3 Path:", model_s3_path)
```

```
Fixed Model S3 Path: s3://sagemaker-ca-central-1-263245924849/airbnb-price-model-f
ixed/model.tar.gz
```

In [55]:
```
model_s3_path = session.upload_data(
    path="model.tar.gz",
    bucket=bucket,
    key_prefix="airbnb-price-model-fixed"
)

print("Fixed Model S3 Path:", model_s3_path)
```

```
Fixed Model S3 Path: s3://sagemaker-ca-central-1-263245924849/airbnb-price-model-f
ixed/model.tar.gz
```

In [56]:
```
%%writefile inference.py

import os
import joblib
import numpy as np

def model_fn(model_dir):
    model_path = os.path.join(model_dir, "model.joblib")
    model = joblib.load(model_path)
```

```
    return model

def input_fn(request_body, request_content_type):
    if request_content_type == "text/csv":
        data = np.array([float(x) for x in request_body.split(",")])
        return data.reshape(1, -1)
    else:
        raise ValueError("Unsupported content type")

def predict_fn(input_data, model):
    prediction = model.predict(input_data)
    return prediction

def output_fn(prediction, content_type):
    return str(prediction[0])
```

```
Overwriting inference.py
```

In [57]:
```
from sagemaker.sklearn.model import SKLearnModel

sk_model = SKLearnModel(
    model_data=model_s3_path,
    role=role,
    entry_point="inference.py",
    framework_version="1.2-1"
)
```

In [ ]:
```
predictor = sk_model.deploy(
    initial_instance_count=1,
    instance_type="ml.t2.medium"
)

print("Endpoint deployed successfully.")
```

In [ ]:
```
print("still alive")
```

In [59]:
```
import sklearn
print(sklearn.__version__)
```

```
1.8.0
```

In [60]:
```
!pip uninstall -y scikit-learn
!pip install scikit-learn==1.2.2
```

```
Found existing installation: scikit-learn 1.8.0
Uninstalling scikit-learn-1.8.0:
  Successfully uninstalled scikit-learn-1.8.0
Collecting scikit-learn==1.2.2
  Downloading scikit-learn-1.2.2.tar.gz (7.3 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 7.3/7.3 MB 61.5 MB/s  0:00:00
  Installing build dependencies ... error
  error: subprocess-exited-with-error

  × installing build dependencies for scikit-learn did not run successfully.
  │ exit code: 1
  ╰─> [81 lines of output]
      Ignoring numpy: markers 'python_version == "3.10" and platform_system == "Wi
ndows" and platform_python_implementation != "PyPy"' don't match your environment
      Collecting setuptools
        Using cached setuptools-82.0.1-py3-none-any.whl.metadata (6.5 kB)
      Collecting wheel
        Using cached wheel-0.46.3-py3-none-any.whl.metadata (2.4 kB)
      Collecting Cython>=0.29.24
        Downloading cython-3.2.4-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x
86_64.manylinux_2_28_x86_64.whl.metadata (7.5 kB)
      Collecting oldest-supported-numpy
        Downloading oldest_supported_numpy-2023.12.21-py3-none-any.whl.metadata
(9.8 kB)
      Collecting scipy>=1.3.2
        Downloading scipy-1.17.1.tar.gz (30.6 MB)
           ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0.0/30.6 MB ? eta -:--:--
           ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 17.8/30.6 MB 109.4 MB/s eta 0:0
0:01
           ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 30.6/30.6 MB 93.2 MB/s  0:00:0
0
        Installing build dependencies: started
        Installing build dependencies: finished with status 'error'
        error: subprocess-exited-with-error

        × installing build dependencies for scipy did not run successfully.
        │ exit code: 1
        ╰─> [55 lines of output]
            Collecting meson-python<0.21.0,>=0.15.0
              Downloading meson_python-0.19.0-py3-none-any.whl.metadata (2.1 kB)
            Collecting Cython<3.3.0,>=3.0.8
              Using cached cython-3.2.4-cp312-cp312-manylinux2014_x86_64.manylinux
_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (7.5 kB)
            Collecting pybind11<3.1.0,>=2.13.2
              Downloading pybind11-3.0.2-py3-none-any.whl.metadata (10 kB)
            Collecting pythran<0.19.0,>=0.14.0
              Downloading pythran-0.18.1-py3-none-any.whl.metadata (7.9 kB)
            Collecting numpy<2.7,>=2.0.0
              Downloading numpy-2.4.4.tar.gz (20.7 MB)
                 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0.0/20.7 MB ? eta -:--:-
-
                 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 19.7/20.7 MB 117.5 MB/s e
ta 0:00:01
                 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 20.7/20.7 MB 91.1 MB/s
0:00:00
              Installing build dependencies: started
```

```
              Installing build dependencies: finished with status 'done'
              Getting requirements to build wheel: started
              Getting requirements to build wheel: finished with status 'done'
              Installing backend dependencies: started
              Installing backend dependencies: finished with status 'done'
              Preparing metadata (pyproject.toml): started
              Preparing metadata (pyproject.toml): finished with status 'error'
              error: subprocess-exited-with-error

              × Preparing metadata (pyproject.toml) did not run successfully.
              │ exit code: 1
              ╰─> [19 lines of output]
                  + /home/ec2-user/anaconda3/envs/python3/bin/python3.12 /tmp/pip-
install-f3cl75ma/numpy_4c3210fd9bb64e328f1c4277910a8076/vendored-meson/meson/meso
n.py setup /tmp/pip-install-f3cl75ma/numpy_4c3210fd9bb64e328f1c4277910a8076 /tmp/p
ip-install-f3cl75ma/numpy_4c3210fd9bb64e328f1c4277910a8076/.mesonpy-rf9yu1wa -Dbui
ldtype=release -Db_ndebug=if-release -Db_vscrt=md --native-file=/tmp/pip-install-f
3cl75ma/numpy_4c3210fd9bb64e328f1c4277910a8076/.mesonpy-rf9yu1wa/meson-python-nati
ve-file.ini
                  The Meson build system
                  Version: 1.9.2
                  Source dir: /tmp/pip-install-f3cl75ma/numpy_4c3210fd9bb64e328f1c
4277910a8076
                  Build dir: /tmp/pip-install-f3cl75ma/numpy_4c3210fd9bb64e328f1c4
277910a8076/.mesonpy-rf9yu1wa
                  Build type: native build
                  Project name: NumPy
                  Project version: 2.4.4
                  C compiler for the host machine: cc (gcc 7.3.1 "cc (GCC) 7.3.1 2
0180712 (Red Hat 7.3.1-18)")
                  C linker for the host machine: cc ld.bfd 2.29.1-31
                  C++ compiler for the host machine: c++ (gcc 7.3.1 "c++ (GCC) 7.
3.1 20180712 (Red Hat 7.3.1-18)")
                  C++ linker for the host machine: c++ ld.bfd 2.29.1-31
                  Cython compiler for the host machine: cython (cython 3.2.4)
                  Host machine cpu family: x86_64
                  Host machine cpu: x86_64

                  ../meson.build:25:4: ERROR: Problem encountered: NumPy requires
GCC >= 9.3

                  A full log can be found at /tmp/pip-install-f3cl75ma/numpy_4c321
0fd9bb64e328f1c4277910a8076/.mesonpy-rf9yu1wa/meson-logs/meson-log.txt
                  [end of output]

              note: This error originates from a subprocess, and is likely not a p
roblem with pip.
            error: metadata-generation-failed

            × Encountered error while generating package metadata.
            ╰─> numpy

            note: This is an issue with the package mentioned above, not pip.
            hint: See above for details.
            [end of output]
```

```
        note: This error originates from a subprocess, and is likely not a problem
with pip.
      ERROR: Failed to build 'scipy' when installing build dependencies for scipy

      [end of output]

  note: This error originates from a subprocess, and is likely not a problem with
pip.
ERROR: Failed to build 'scikit-learn' when installing build dependencies for sciki
t-learn
```

In [61]:
```
!pip install --upgrade pip setuptools wheel
!pip install scikit-learn==1.4.2
```

```
Requirement already satisfied: pip in /home/ec2-user/anaconda3/envs/python3/lib/py
thon3.12/site-packages (26.0.1)
Requirement already satisfied: setuptools in /home/ec2-user/anaconda3/envs/python
3/lib/python3.12/site-packages (81.0.0)
Collecting setuptools
  Using cached setuptools-82.0.1-py3-none-any.whl.metadata (6.5 kB)
Requirement already satisfied: wheel in /home/ec2-user/anaconda3/envs/python3/lib/
python3.12/site-packages (0.46.3)
Requirement already satisfied: packaging>=24.0 in /home/ec2-user/anaconda3/envs/py
thon3/lib/python3.12/site-packages (from wheel) (24.2)
Using cached setuptools-82.0.1-py3-none-any.whl (1.0 MB)
Installing collected packages: setuptools
  Attempting uninstall: setuptools
    Found existing installation: setuptools 81.0.0
    Uninstalling setuptools-81.0.0:
      Successfully uninstalled setuptools-81.0.0
ERROR: pip's dependency resolver does not currently take into account all the pack
ages that are installed. This behaviour is the source of the following dependency
conflicts.
jupyterlab 4.5.3 requires httpx<1,>=0.25.0, which is not installed.
Successfully installed setuptools-82.0.1
Collecting scikit-learn==1.4.2
  Downloading scikit_learn-1.4.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x
86_64.whl.metadata (11 kB)
Requirement already satisfied: numpy>=1.19.5 in /home/ec2-user/anaconda3/envs/pyth
on3/lib/python3.12/site-packages (from scikit-learn==1.4.2) (1.26.4)
Requirement already satisfied: scipy>=1.6.0 in /home/ec2-user/anaconda3/envs/pytho
n3/lib/python3.12/site-packages (from scikit-learn==1.4.2) (1.17.0)
Requirement already satisfied: joblib>=1.2.0 in /home/ec2-user/anaconda3/envs/pyth
on3/lib/python3.12/site-packages (from scikit-learn==1.4.2) (1.5.3)
Requirement already satisfied: threadpoolctl>=2.0.0 in /home/ec2-user/anaconda3/en
vs/python3/lib/python3.12/site-packages (from scikit-learn==1.4.2) (3.6.0)
Downloading scikit_learn-1.4.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86
_64.whl (12.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.2/12.2 MB 83.9 MB/s  0:00:00
Installing collected packages: scikit-learn
ERROR: pip's dependency resolver does not currently take into account all the pack
ages that are installed. This behaviour is the source of the following dependency
conflicts.
shap 0.50.0 requires numpy>=2, but you have numpy 1.26.4 which is incompatible.
Successfully installed scikit-learn-1.4.2
```

In [62]:
```
import sklearn
print(sklearn.__version__)
```

```
1.8.0
```

In [1]:
```
import sklearn
print(sklearn.__version__)
```

```
1.4.2
```

In [5]:
```
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
import pandas as pd

# Load clean training data
train_df = pd.read_csv("sagemaker_train.csv", header=None)

X = train_df.iloc[:, 1:]
y = train_df.iloc[:, 0]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

rf_model = RandomForestRegressor(random_state=42)
rf_model.fit(X_train, y_train)

print("Model retrained successfully.")
```

```
Model retrained successfully.
```

In [6]:
```
import joblib

joblib.dump(rf_model, "model.joblib")
print("Retrained model saved.")
```

```
Retrained model saved.
```

In [7]:
```
import tarfile

with tarfile.open("model.tar.gz", "w:gz") as tar:
    tar.add("model.joblib")

print("Updated model.tar.gz created.")
```

```
Updated model.tar.gz created.
```

In [9]:
```
import sagemaker
from sagemaker import get_execution_role

session = sagemaker.Session()
bucket = session.default_bucket()
role = get_execution_role()

print("Bucket:", bucket)
print("Role:", role)
```

```
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemake
r/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.c
onfig/sagemaker/config.yaml
Bucket: sagemaker-ca-central-1-263245924849
Role: arn:aws:iam::263245924849:role/service-role/AmazonSageMaker-ExecutionRole-20
260323T193815
```

In [10]:
```
model_s3_path = session.upload_data(
    path="model.tar.gz",
    bucket=bucket,
    key_prefix="airbnb-price-model-v142"
)

print("New model uploaded:", model_s3_path)
```

```
New model uploaded: s3://sagemaker-ca-central-1-263245924849/airbnb-price-model-v1
42/model.tar.gz
```

In [11]:
```
from sagemaker.sklearn.model import SKLearnModel

sk_model = SKLearnModel(
    model_data=model_s3_path,
    role=role,
    entry_point="inference.py",
    framework_version="1.2-1"
)
```

In [12]:
```
from sagemaker.sklearn.model import SKLearnModel

sk_model = SKLearnModel(
    model_data=model_s3_path,
    role=role,
    entry_point="inference.py",
    framework_version="1.2-1"
)
```

In [13]:
```
predictor = sk_model.deploy(
    initial_instance_count=1,
    instance_type="ml.t2.medium"
)

print("Endpoint deployed successfully.")
```

```
-------------------------------------------------------*
Please check the troubleshooting guide for common errors: https://docs.aws.amazon.
com/sagemaker/latest/dg/sagemaker-python-sdk-troubleshooting.html#sagemaker-python
-sdk-troubleshooting-create-endpoint
```

```
Traceback (most recent call last)
in <module>:1

❯ 1 predictor = sk_model.deploy(
  2 │   initial_instance_count=1,
  3 │   instance_type="ml.t2.medium"
  4 )

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/sagemaker
deploy

  1822 │   │   │   │   )
  1823 │   │   │   │   self.sagemaker_session.update_endpoint(self.endpoint_
  1824 │   │   │   else:
❯ 1825 │   │   │   │   self.sagemaker_session.endpoint_from_production_varia
  1826 │   │   │   │   │   name=self.endpoint_name,
  1827 │   │   │   │   │   production_variants=[production_variant],
  1828 │   │   │   │   │   tags=tags,

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/sagemaker
endpoint_from_production_variants

  6251 │   │   logger.info("Creating endpoint-config with name %s", name)
  6252 │   │   self.sagemaker_client.create_endpoint_config(**config_options
  6253 │   │
❯ 6254 │   │   return self.create_endpoint(
  6255 │   │   │   endpoint_name=name,
  6256 │   │   │   config_name=name,
  6257 │   │   │   tags=endpoint_tags,

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/sagemaker
create_endpoint

  5096 │   │   │   logger.error(
  5097 │   │   │   │   "Please check the troubleshooting guide for common er
  5098 │   │   │   )
❯ 5099 │   │   │   raise e
  5100 │
  5101 │   def endpoint_in_service_or_not(self, endpoint_name: str):
  5102 │       """Check whether an Amazon SageMaker ``Endpoint``` is in IN_S

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/sagemaker
create_endpoint

  5085 │   │   │   │   self.endpoint_arn = res["EndpointArn"]
  5086 │   │   │
  5087 │   │   │   if wait:
❯ 5088 │   │   │   │   self.wait_for_endpoint(endpoint_name, live_logging=li
  5089 │   │   │   return endpoint_name
  5090 │   │   except Exception as e:
  5091 │   │   │   troubleshooting = (

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/sagemaker
wait_for_endpoint
```

```
5894 |   |   |   |   |   allowed_statuses=["InService"],
5895 |   |   |   |   |   actual_status=status,
5896 |   |   |   |   )
❯ 5897 |   |   |   raise exceptions.UnexpectedStatusException(
5898 |   |   |   |   message=message,
5899 |   |   |   |   allowed_statuses=["InService"],
5900 |   |   |   |   actual_status=status,
```

UnexpectedStatusException: Error hosting endpoint sagemaker-scikit-learn-2026-
The primary container for production variant AllTraffic did not pass the ping
logs for this endpoint.. Try changing the instance type or reference the troub

In [1]:
```
from sagemaker.sklearn.model import SKLearnModel

sk_model = SKLearnModel(
    model_data=model_s3_path,
    role=role,
    entry_point="inference.py",
    framework_version="1.4"   # ← change this from "1.2-1"
)

predictor = sk_model.deploy(
    initial_instance_count=1,
    instance_type="ml.t2.medium"
)
print("Endpoint deployed successfully.")
```

```
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemake
r/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.c
onfig/sagemaker/config.yaml
```

Traceback (most recent call last)

```
in <module>:4

    1 from sagemaker.sklearn.model import SKLearnModel
    2
    3 sk_model = SKLearnModel(
❯   4 |   model_data=model_s3_path,
    5 |   role=role,
    6 |   entry_point="inference.py",
    7 |   framework_version="1.4"   # ← change this from "1.2-1"
```

NameError: name 'model_s3_path' is not defined

In [2]:
```
import pandas as pd
import joblib
import tarfile
import os
import sagemaker
from sagemaker import get_execution_role
from sagemaker.sklearn.model import SKLearnModel
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
```

```python
# 1. Load and prep data
train_df = pd.read_csv("sagemaker_train.csv", header=None)
X = train_df.iloc[:, 1:]
y = train_df.iloc[:, 0]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 2. Train model
rf_model = RandomForestRegressor(random_state=42)
rf_model.fit(X_train, y_train)
print("✅ Model trained")

# 3. Save model
joblib.dump(rf_model, "model.joblib")
print("✅ model.joblib saved")

# 4. Package into tar.gz
with tarfile.open("model.tar.gz", "w:gz") as tar:
    tar.add("model.joblib")
print("✅ model.tar.gz created")

# 5. SageMaker session
session = sagemaker.Session()
bucket = session.default_bucket()
role = get_execution_role()
print("Bucket:", bucket)

# 6. Upload to S3
model_s3_path = session.upload_data(
    path="model.tar.gz",
    bucket=bucket,
    key_prefix="airbnb-price-final"
)
print("✅ Uploaded to S3:", model_s3_path)

# 7. Deploy
sk_model = SKLearnModel(
    model_data=model_s3_path,
    role=role,
    entry_point="inference.py",
    framework_version="1.4"
)

predictor = sk_model.deploy(
    initial_instance_count=1,
    instance_type="ml.m5.large"   # more reliable than t2.medium
)
print("✅ Endpoint deployed successfully!")
```

```
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemake
r/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.c
onfig/sagemaker/config.yaml
✅ Model trained
✅ model.joblib saved
✅ model.tar.gz created
Bucket: sagemaker-ca-central-1-263245924849
✅ Uploaded to S3: s3://sagemaker-ca-central-1-263245924849/airbnb-price-final/mod
el.tar.gz
```

```
Traceback (most recent call last)
in <module>:56

  53 │   framework_version="1.4"
  54 )
  55
❯ 56 predictor = sk_model.deploy(
  57 │   initial_instance_count=1,
  58 │   instance_type="ml.m5.large"   # more reliable than t2.medium
  59 )

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/sagemaker
deploy

  1745 │   │   │   return None
  1746 │   │
  1747 │   │   else:  # existing single model endpoint path
❯ 1748 │   │   │   self._create_sagemaker_model(
  1749 │   │   │   │   instance_type=instance_type,
  1750 │   │   │   │   accelerator_type=accelerator_type,
  1751 │   │   │   │   tags=tags,

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/sagemaker
_create_sagemaker_model

   944 │   │   │   if self.name is None and model_package.name is not None:
   945 │   │   │   │   self.name = model_package.name
   946 │   │   else:
❯  947 │   │   │   container_def = self.prepare_container_def(
   948 │   │   │   │   instance_type,
   949 │   │   │   │   accelerator_type=accelerator_type,
   950 │   │   │   │   serverless_inference_config=serverless_inference_conf

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/sagemaker
2 in prepare_container_def

  319 │   │
  320 │   │   deploy_image = self.image_uri
  321 │   │   if not deploy_image:
❯ 322 │   │   │   deploy_image = self.serving_image_uri(
  323 │   │   │   │   self.sagemaker_session.boto_region_name, instance_type
  324 │   │   │   )
  325

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/sagemaker
1 in serving_image_uri

  358 │   │   │   str: The appropriate image URI based on the given paramete
  359 │   │
  360 │   │   """
❯ 361 │   │   return image_uris.retrieve(
  362 │   │   │   self._framework_name,
  363 │   │   │   region_name,
  364 │   │   │   version=self.framework_version,
```

```
/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/sagemaker
py:432 in wrapper

  429 │   │   │   if isinstance(value, Parameter):
  430 │   │   │   │   logger.warning(warning_msg_template, arg_name, func_na
  431 │   │   │   │   kwargs[arg_name] = value.default_value
❯ 432 │   │   return func(*args, **kwargs)
  433 │
  434 │   return wrapper
  435

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/sagemaker
in retrieve

  199 │   │   config = _config_for_framework_and_scope(_framework, final_ima
  200 │
  201 │   original_version = version
❯ 202 │   version = _validate_version_and_set_if_needed(version, config, fra
  203 │   version_config = config["versions"][_version_for_config(version, c
  204 │
  205 │   if framework == HUGGING_FACE_FRAMEWORK:

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/sagemaker
in _validate_version_and_set_if_needed

  510 │   │   return available_versions[0]
  511 │   if not version:
  512 │   │   version = _get_latest_version(framework, version, image_scope)
❯ 513 │   _validate_arg(version, available_versions + aliased_versions, "{}_
  514 │   return version
  515
  516

/home/ec2-user/anaconda3/envs/python3/lib/python3.12/site-packages/sagemaker
in _validate_arg

  622 def _validate_arg(arg, available_options, arg_name):
  623 │   """Checks if the arg is in the available options, and raises a ``V
  624 │   if arg not in available_options:
❯ 625 │   │   raise ValueError(
  626 │   │   │   "Unsupported {arg_name}: {arg}. You may need to upgrade yo
  627 │   │   │   "(pip install -U sagemaker) for newer {arg_name}s. Support
  628 │   │   │   "{options}.".format(arg_name=arg_name, arg=arg, options=",
```

ValueError: Unsupported sklearn version: **1.4**. You may need to upgrade your SDK

In [3]:

```python
sk_model = SKLearnModel(
    model_data=model_s3_path,
    role=role,
    entry_point="inference.py",
    framework_version="1.4-2"  # ← was "1.4", must be exact: "1.4-2"
)

predictor = sk_model.deploy(
```

```
    initial_instance_count=1,
    instance_type="ml.m5.large"
)
print("✅ Endpoint deployed successfully!")
```

```
-----!✅ Endpoint deployed successfully!
```

In [4]:

```
import numpy as np
from sagemaker.serializers import CSVSerializer
from sagemaker.deserializers import StringDeserializer

# Set serializer/deserializer
predictor.serializer = CSVSerializer()
predictor.deserializer = StringDeserializer()

# Grab a sample row from test data (without the price label)
sample = X_test.iloc[0].values
sample_csv = ",".join(str(x) for x in sample)

# Run prediction
result = predictor.predict(sample_csv)

print("Predicted Price: $", result)
print("Actual Price:    $", y_test.iloc[0])
```

```
Predicted Price: $ 182.37
Actual Price:    $ 346
```

In [5]:

```
from sklearn.metrics import mean_absolute_error, r2_score, mean_squared_error
import numpy as np

# Predict on entire test set
all_predictions = []
for i in range(len(X_test)):
    sample = X_test.iloc[i].values
    sample_csv = ",".join(str(x) for x in sample)
    pred = predictor.predict(sample_csv)
    all_predictions.append(float(pred))

y_pred = np.array(all_predictions)
y_actual = y_test.values

# Metrics
mae = mean_absolute_error(y_actual, y_pred)
rmse = np.sqrt(mean_squared_error(y_actual, y_pred))
r2 = r2_score(y_actual, y_pred)

print("=" * 40)
print("📊 Endpoint Evaluation Results")
print("=" * 40)
print(f"MAE:  ${mae:.2f}")
print(f"RMSE: ${rmse:.2f}")
print(f"R²:   {r2:.4f}")
print("=" * 40)
```

```
========================================
📊 Endpoint Evaluation Results
========================================
MAE:  $74.70
RMSE: $126.17
R²:   0.2725
========================================
```

In [6]:
```
predictor.delete_endpoint()
print("✅ Endpoint deleted. No more charges.")
```

```
✅ Endpoint deleted. No more charges.
```

In [7]:
```
import pandas as pd
import numpy as np

df_raw = pd.read_csv("airbnb.csv")
print(df_raw.shape)
print(df_raw.dtypes)
print(df_raw.head())
```

```
(953, 7)
Title                     object
Detail                    object
Date                      object
Price(in dollar)          object
Offer price(in dollar)    object
Review and rating         object
Number of bed             object
dtype: object
                                Title  \
0  Chalet in Skykomish, Washington, US
1       Cabin in Hancock, New York, US
2   Cabin in West Farmington, Ohio, US
3      Home in Blue Ridge, Georgia, US
4    Treehouse in Grandview, Texas, US

                                            Detail         Date  \
0                          Sky Haus - A-Frame Cabin  Jun 11 - 16
1    The Catskill A-Frame - Mid-Century Modern Cabin   Jun 6 - 11
2  The Triangle: A-Frame Cabin for your city retreat   Jul 9 - 14
3  *Summer Sizzle* 5 Min to Blue Ridge* Pets* Hot...  Jun 11 - 16
4  Luxury Treehouse Couples Getaway w/ Peaceful V...    Jun 4 - 9

  Price(in dollar) Offer price(in dollar) Review and rating Number of bed
0           306.00                 229.00        4.85 (531)        4 beds
1           485.00                 170.00        4.77 (146)        4 beds
2           119.00                 522.00        4.91 (515)        4 beds
3           192.00                 348.00         4.94 (88)        5 beds
4           232.00                 196.00        4.99 (222)   1 queen bed
```

In [9]:
```
import pandas as pd
import numpy as np
import re

df_raw = pd.read_csv("airbnb.csv")
```

```
# 1. Target variable - clean price
df_raw['price'] = pd.to_numeric(df_raw['Price(in dollar)'], errors='coerce')

# 2. Offer price & discount
df_raw['offer_price'] = pd.to_numeric(df_raw['Offer price(in dollar)'], errors='coe
df_raw['discount'] = df_raw['price'] - df_raw['offer_price']
df_raw['discount_pct'] = df_raw['discount'] / df_raw['price']

# 3. Rating and number of reviews
df_raw['rating'] = df_raw['Review and rating'].str.extract(r'(\d+\.\d+)').astype(fl
df_raw['num_reviews'] = df_raw['Review and rating'].str.extract(r'\((\d+)\)').astyp

# 4. Number of beds
df_raw['beds'] = df_raw['Number of bed'].str.extract(r'(\d+)').astype(float)
df_raw['is_queen'] = df_raw['Number of bed'].str.contains('queen', case=False).asty
df_raw['is_king'] = df_raw['Number of bed'].str.contains('king', case=False).astype

# 5. Location features from Title
df_raw['country'] = df_raw['Title'].str.extract(r',\s*(\w+)$')
df_raw['state'] = df_raw['Title'].str.extract(r',\s*(\w+),\s*\w+$')
df_raw['property_type'] = df_raw['Title'].str.extract(r'^(\w+)\s+in')

# 6. Date/season features
df_raw['month'] = pd.to_datetime(df_raw['Date'].str.split(' - ').str[0] + ' 2024',
                                 format='%b %d %Y').dt.month
df_raw['is_summer'] = df_raw['month'].isin([6, 7, 8]).astype(int)
df_raw['is_weekend_checkin'] = df_raw['Date'].str.extract(r'(\d+)').astype(float) %

# 7. Text features from Detail
df_raw['detail_length'] = df_raw['Detail'].str.len()
df_raw['is_luxury'] = df_raw['Detail'].str.contains('luxury|Luxury', na=False).asty
df_raw['has_hot_tub'] = df_raw['Detail'].str.contains('hot tub|Hot tub', na=False).
df_raw['has_pool'] = df_raw['Detail'].str.contains('pool|Pool', na=False).astype(in
df_raw['has_pets'] = df_raw['Detail'].str.contains('pet|Pet', na=False).astype(int)
df_raw['has_wifi'] = df_raw['Detail'].str.contains('wifi|WiFi|Wifi', na=False).asty
df_raw['is_cabin'] = df_raw['Detail'].str.contains('cabin|Cabin', na=False).astype(
df_raw['is_treehouse'] = df_raw['Detail'].str.contains('treehouse|Treehouse', na=Fa

print(df_raw.shape)
print(df_raw[['price', 'offer_price', 'discount_pct', 'rating', 'num_reviews',
              'beds', 'state', 'property_type', 'month', 'is_luxury',
              'has_hot_tub', 'has_pool']].head(10))
```

```
(953, 30)
   price  offer_price  discount_pct  rating  num_reviews  beds       state  \
0  306.0        229.0      0.251634    4.85        531.0   4.0  Washington
1  485.0        170.0      0.649485    4.77        146.0   4.0         NaN
2  119.0        522.0     -3.386555    4.91        515.0   4.0        Ohio
3  192.0        348.0     -0.812500    4.94         88.0   5.0     Georgia
4  232.0        196.0      0.155172    4.99        222.0   1.0       Texas
5  261.0        148.0      0.432950    4.84        555.0   1.0         NaN
6  607.0          NaN           NaN    4.64         90.0   2.0       Maine
7  381.0          NaN           NaN    5.00         48.0   2.0         NaN
8   29.0          NaN           NaN    4.84         58.0   2.0         NaN
9  154.0          NaN           NaN    4.92        744.0   3.0      Nevada

  property_type  month  is_luxury  has_hot_tub  has_pool
0        Chalet      6          0            0         0
1         Cabin      6          0            0         0
2         Cabin      7          0            0         0
3          Home      6          0            1         0
4     Treehouse      6          1            0         0
5           NaN      6          0            0         0
6          Home      6          0            0         0
7        Chalet      6          0            0         0
8          Room      7          0            0         0
9           NaN      6          0            0         0
```

In [10]:

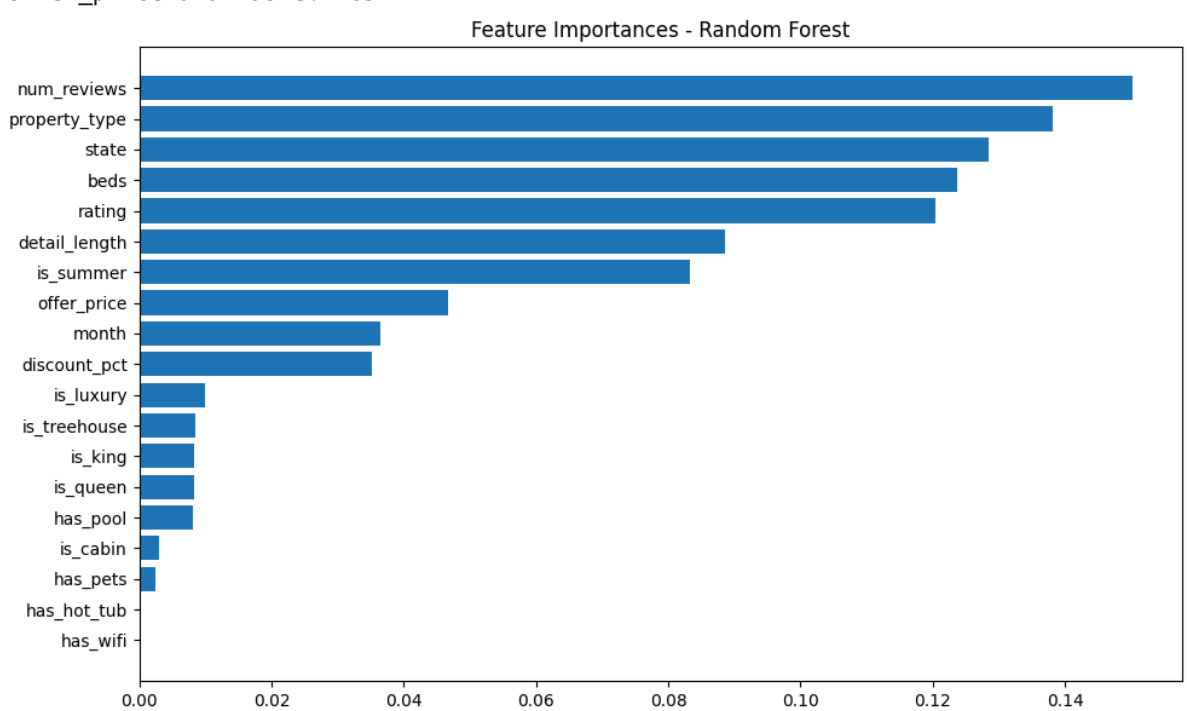
```python
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import mean_absolute_error, r2_score, mean_squared_error

# 1. Select final features
features = [
    'offer_price', 'discount_pct', 'rating', 'num_reviews',
    'beds', 'is_queen', 'is_king', 'month', 'is_summer',
    'is_luxury', 'has_hot_tub', 'has_pool', 'has_pets',
    'has_wifi', 'is_cabin', 'is_treehouse', 'detail_length',
    'state', 'property_type'
]

df_model = df_raw[features + ['price']].copy()

# 2. Fix discount_pct (negative = offer > price, likely data error)
df_model['discount_pct'] = df_model['discount_pct'].clip(0, 1)

# 3. Encode categorical columns
for col in ['state', 'property_type']:
    df_model[col] = df_model[col].fillna('Unknown')
    le = LabelEncoder()
    df_model[col] = le.fit_transform(df_model[col])

# 4. Fill remaining NaNs with median
df_model = df_model.fillna(df_model.median(numeric_only=True))

# 5. Drop rows where price is missing
df_model = df_model.dropna(subset=['price'])
```

```python
print("Dataset shape:", df_model.shape)
print("NaNs remaining:", df_model.isnull().sum().sum())

# 6. Split
X = df_model.drop('price', axis=1)
y = df_model['price']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 7. Train Random Forest
rf = RandomForestRegressor(n_estimators=300, max_depth=10,
                           min_samples_leaf=3, random_state=42)
rf.fit(X_train, y_train)
rf_pred = rf.predict(X_test)

# 8. Train Gradient Boosting
gb = GradientBoostingRegressor(n_estimators=300, max_depth=5,
                               learning_rate=0.05, random_state=42)
gb.fit(X_train, y_train)
gb_pred = gb.predict(X_test)

# 9. Compare results
print("\n" + "="*45)
print("📊 Model Comparison")
print("="*45)
for name, pred in [("Random Forest", rf_pred), ("Gradient Boosting", gb_pred)]:
    mae = mean_absolute_error(y_test, pred)
    rmse = np.sqrt(mean_squared_error(y_test, pred))
    r2 = r2_score(y_test, pred)
    print(f"\n{name}:")
    print(f"  MAE:  ${mae:.2f}")
    print(f"  RMSE: ${rmse:.2f}")
    print(f"  R²:   {r2:.4f}")
print("="*45)
```

```
Dataset shape: (953, 20)
NaNs remaining: 0

=============================================
📊 Model Comparison
=============================================

Random Forest:
  MAE:  $78.57
  RMSE: $123.38
  R²:   0.1599

Gradient Boosting:
  MAE:  $74.51
  RMSE: $118.14
  R²:   0.2297
=============================================
```

In [11]:
```
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Check how many NaNs offer_price had originally
print("offer_price NaN count:", df_raw['offer_price'].isna().sum())
print("offer_price available:", df_raw['offer_price'].notna().sum())

# 2. Feature importance - what's actually driving predictions?
feature_names = X.columns.tolist()
importances = rf.feature_importances_

plt.figure(figsize=(10, 6))
sorted_idx = np.argsort(importances)
plt.barh([feature_names[i] for i in sorted_idx], importances[sorted_idx])
plt.title("Feature Importances - Random Forest")
plt.tight_layout()
plt.show()

# 3. Check correlation of each feature with price
print("\n📊 Correlation with price:")
corr = df_model.corr()['price'].sort_values(ascending=False)
print(corr)
```

```
offer_price NaN count: 788
offer_price available: 165
```



```
📊 Correlation with price:
price            1.000000
beds             0.234489
rating           0.180248
discount_pct     0.152438
offer_price      0.105611
is_luxury        0.102183
detail_length    0.078130
is_treehouse     0.058305
has_pool         0.049314
property_type    0.041055
has_hot_tub      0.039617
has_pets         0.021422
is_cabin        -0.001267
is_queen        -0.001399
month           -0.016199
is_king         -0.034715
num_reviews     -0.044905
has_wifi        -0.058044
state           -0.170598
is_summer       -0.197000
Name: price, dtype: float64
```

In [12]:

```
# Drop weak/noisy features, keep only meaningful ones
features_v2 = [
    'beds',           # 0.23 - strongest
    'rating',         # 0.18
    'discount_pct',   # 0.15
    'is_luxury',      # 0.10
    'detail_length',  # 0.08
    'is_treehouse',   # 0.06
    'has_pool',       # 0.05
    'has_hot_tub',    # 0.04
    'has_pets',       # 0.02
    'property_type',  # 0.04 - keep as it has real categories
    'num_reviews',    # weak but keep
]

# Rebuild WITHOUT offer_price, state, is_summer, is_king, is_queen, month
df_model2 = df_raw[features_v2 + ['price']].copy()

# Re-encode property_type properly (one-hot instead of label encode)
df_model2 = pd.get_dummies(df_model2, columns=['property_type'])

# Fill NaNs
df_model2 = df_model2.fillna(df_model2.median(numeric_only=True))
df_model2 = df_model2.dropna(subset=['price'])

print("Shape:", df_model2.shape)
print("Columns:", df_model2.columns.tolist())

X2 = df_model2.drop('price', axis=1)
y2 = df_model2['price']

X2_train, X2_test, y2_train, y2_test = train_test_split(
    X2, y2, test_size=0.2, random_state=42
```

```
)

# Train both models
rf2 = RandomForestRegressor(n_estimators=300, max_depth=8,
                            min_samples_leaf=2, random_state=42)
rf2.fit(X2_train, y2_train)
rf2_pred = rf2.predict(X2_test)

gb2 = GradientBoostingRegressor(n_estimators=300, max_depth=4,
                                learning_rate=0.05, random_state=42)
gb2.fit(X2_train, y2_train)
gb2_pred = gb2.predict(X2_test)

print("\n" + "="*45)
print("📊 Improved Model Comparison")
print("="*45)
for name, pred in [("Random Forest v2", rf2_pred), ("Gradient Boosting v2", gb2_pre
    mae = mean_absolute_error(y2_test, pred)
    rmse = np.sqrt(mean_squared_error(y2_test, pred))
    r2 = r2_score(y2_test, pred)
    print(f"\n{name}:")
    print(f"  MAE:  ${mae:.2f}")
    print(f"  RMSE: ${rmse:.2f}")
    print(f"  R²:   {r2:.4f}")
print("="*45)
```

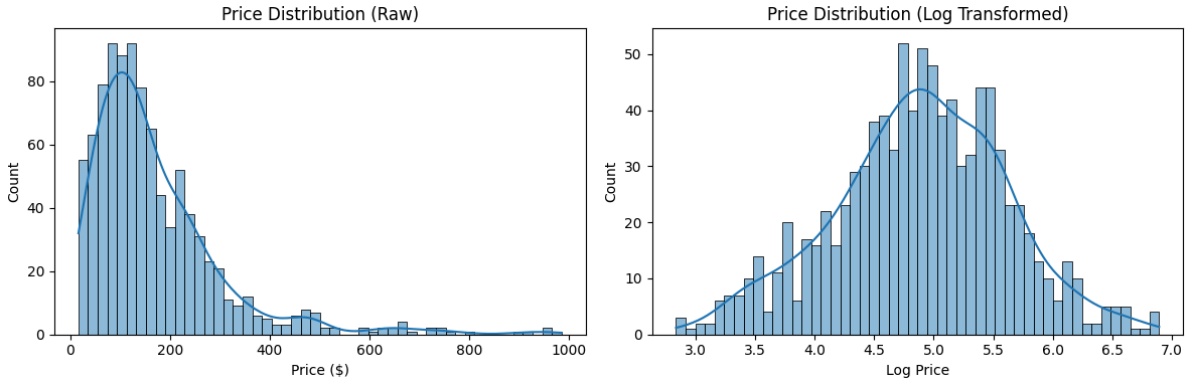
```
Shape: (953, 42)
Columns: ['beds', 'rating', 'discount_pct', 'is_luxury', 'detail_length', 'is_tree
house', 'has_pool', 'has_hot_tub', 'has_pets', 'num_reviews', 'price', 'property_t
ype_Apartment', 'property_type_Barn', 'property_type_Boat', 'property_type_Bungalo
w', 'property_type_Cabin', 'property_type_Campsite', 'property_type_Castle', 'prop
erty_type_Cave', 'property_type_Chalet', 'property_type_Condo', 'property_type_Cot
tage', 'property_type_Dome', 'property_type_Guesthouse', 'property_type_Home', 'pr
operty_type_Hostel', 'property_type_Hotel', 'property_type_Houseboat', 'property_t
ype_Hut', 'property_type_Island', 'property_type_Lighthouse', 'property_type_Lof
t', 'property_type_Resort', 'property_type_Room', 'property_type_Tent', 'property_
type_Tower', 'property_type_Townhouse', 'property_type_Treehouse', 'property_type_
Trullo', 'property_type_Villa', 'property_type_Windmill', 'property_type_Yurt']

=============================================
📊 Improved Model Comparison
=============================================

Random Forest v2:
  MAE:  $79.23
  RMSE: $122.36
  R²:   0.1736

Gradient Boosting v2:
  MAE:  $78.70
  RMSE: $123.03
  R²:   0.1645
=============================================
```

In [13]:

```
print("📊 Price Distribution:")
print(df_raw['price'].describe())
```

```
print(f"\nPrices above $500: {(df_raw['price'] > 500).sum()}")
print(f"Prices above $800: {(df_raw['price'] > 800).sum()}")

# Plot distribution
plt.figure(figsize=(12, 4))

plt.subplot(1, 2, 1)
sns.histplot(df_raw['price'], bins=50, kde=True)
plt.title("Price Distribution (Raw)")
plt.xlabel("Price ($)")

plt.subplot(1, 2, 2)
sns.histplot(np.log1p(df_raw['price']), bins=50, kde=True)
plt.title("Price Distribution (Log Transformed)")
plt.xlabel("Log Price")

plt.tight_layout()
plt.show()

# Check outliers
q99 = df_raw['price'].quantile(0.99)
q95 = df_raw['price'].quantile(0.95)
q01 = df_raw['price'].quantile(0.01)
print(f"\n1st percentile:  ${q01:.0f}")
print(f"95th percentile: ${q95:.0f}")
print(f"99th percentile: ${q99:.0f}")
```

```
📊 Price Distribution:
count    952.000000
mean     170.000000
std      135.398784
min       16.000000
25%       82.750000
50%      134.500000
75%      220.250000
max      986.000000
Name: price, dtype: float64

Prices above $500: 27
Prices above $800: 5
```




```
1st percentile:  $24
95th percentile: $441
99th percentile: $727
```

In [14]:
```
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.metrics import mean_absolute_error, r2_score, mean_squared_error

# 1. Remove outliers - cap at 95th percentile ($441)
df_clean = df_model2[df_model2['price'] <= 441].copy()
print(f"Rows after removing outliers: {len(df_clean)} (removed {len(df_model2) - le

# 2. Log transform price
df_clean['log_price'] = np.log1p(df_clean['price'])

X3 = df_clean.drop(['price', 'log_price'], axis=1)
y3_raw = df_clean['price']
y3_log = df_clean['log_price']

X3_train, X3_test, y3_train_log, y3_test_log = train_test_split(
    X3, y3_log, test_size=0.2, random_state=42
)
_, _, y3_train_raw, y3_test_raw = train_test_split(
    X3, y3_raw, test_size=0.2, random_state=42
)

# 3. Train on log price
rf3 = RandomForestRegressor(n_estimators=300, max_depth=8,
                            min_samples_leaf=2, random_state=42)
rf3.fit(X3_train, y3_train_log)

gb3 = GradientBoostingRegressor(n_estimators=300, max_depth=4,
                                learning_rate=0.05, random_state=42)
gb3.fit(X3_train, y3_train_log)

print("\n" + "="*45)
print("📊 Outlier Removed + Log Transform")
print("="*45)
for name, model in [("Random Forest v3", rf3), ("Gradient Boosting v3", gb3)]:
    # Predict in log space, convert back
    pred_log = model.predict(X3_test)
    pred = np.expm1(pred_log)

    mae = mean_absolute_error(y3_test_raw, pred)
    rmse = np.sqrt(mean_squared_error(y3_test_raw, pred))
    r2 = r2_score(y3_test_raw, pred)
    print(f"\n{name}:")
    print(f"  MAE:  ${mae:.2f}")
    print(f"  RMSE: ${rmse:.2f}")
    print(f"  R²:   {r2:.4f}")
print("="*45)
```

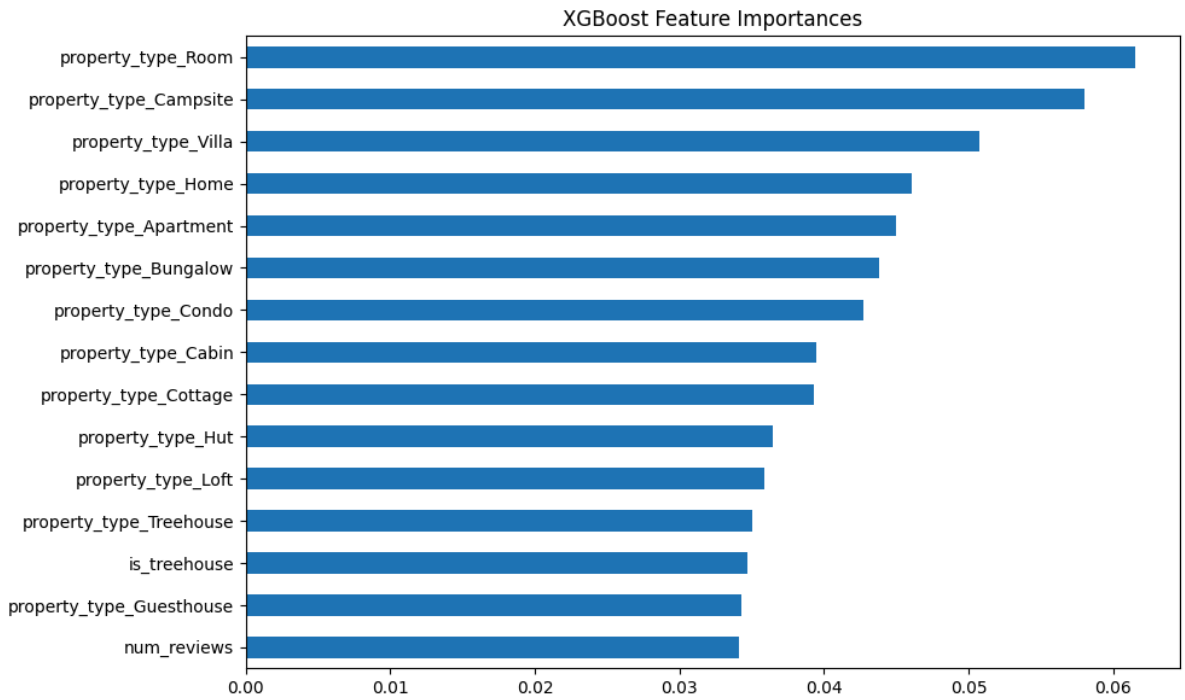
```
Rows after removing outliers: 905 (removed 48)

============================================
📊 Outlier Removed + Log Transform
============================================

Random Forest v3:
  MAE:  $55.56
  RMSE: $74.84
  R²:   0.1980

Gradient Boosting v3:
  MAE:  $57.82
  RMSE: $77.48
  R²:   0.1402
============================================
```

In [16]:
```
!pip install xgboost==1.7.6 -q
```

In [17]:
```
!pip install xgboost -q

from xgboost import XGBRegressor
from sklearn.model_selection import cross_val_score, GridSearchCV

# 1. Try XGBoost
xgb = XGBRegressor(
    n_estimators=500,
    max_depth=4,
    learning_rate=0.03,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=42,
    verbosity=0
)
xgb.fit(X3_train, y3_train_log)
xgb_pred = np.expm1(xgb.predict(X3_test))

mae = mean_absolute_error(y3_test_raw, xgb_pred)
rmse = np.sqrt(mean_squared_error(y3_test_raw, xgb_pred))
r2 = r2_score(y3_test_raw, xgb_pred)

print("="*45)
print("📊 XGBoost Results")
print("="*45)
print(f"MAE:  ${mae:.2f}")
print(f"RMSE: ${rmse:.2f}")
print(f"R²:   {r2:.4f}")
print("="*45)

# 2. Cross validation score (more reliable than single split)
cv_scores = cross_val_score(xgb, X3, y3_log, cv=5, scoring='r2')
print(f"\n5-Fold CV R² scores: {cv_scores.round(3)}")
print(f"Mean CV R²: {cv_scores.mean():.4f} (+/- {cv_scores.std():.4f})")

# 3. Feature importance
import matplotlib.pyplot as plt
```

```
feat_imp = pd.Series(xgb.feature_importances_, index=X3.columns)
feat_imp.sort_values().tail(15).plot(kind='barh', figsize=(10, 6))
plt.title("XGBoost Feature Importances")
plt.tight_layout()
plt.show()
```

```
============================================
📊 XGBoost Results
============================================
MAE:  $57.99
RMSE: $76.98
R²:   0.1513
============================================

5-Fold CV R² scores: [ 0.151 -0.111 -0.015 -0.03  -1.232]
Mean CV R²: -0.2476 (+/- 0.4994)
```



In [18]:

```
from sklearn.ensemble import HistGradientBoostingRegressor
from sklearn.model_selection import cross_val_score

# HistGradientBoosting - sklearn's modern boosting, no install needed
hgb = HistGradientBoostingRegressor(
    max_iter=500,
    max_depth=4,
    learning_rate=0.03,
    min_samples_leaf=10,
    random_state=42
)
hgb.fit(X3_train, y3_train_log)
hgb_pred = np.expm1(hgb.predict(X3_test))

mae = mean_absolute_error(y3_test_raw, hgb_pred)
rmse = np.sqrt(mean_squared_error(y3_test_raw, hgb_pred))
r2 = r2_score(y3_test_raw, hgb_pred)
```

```
print("="*45)
print("📊 HistGradientBoosting Results")
print("="*45)
print(f"MAE:  ${mae:.2f}")
print(f"RMSE: ${rmse:.2f}")
print(f"R²:   {r2:.4f}")
print("="*45)

# Cross validation - more honest on small dataset
cv_scores = cross_val_score(hgb, X3, y3_log, cv=5, scoring='r2')
print(f"\n5-Fold CV R² scores: {cv_scores.round(3)}")
print(f"Mean CV R²:  {cv_scores.mean():.4f}")
print(f"Std CV R²:   {cv_scores.std():.4f}")
```

```
============================================
📊 HistGradientBoosting Results
============================================
MAE:  $59.13
RMSE: $77.80
R²:   0.1331
============================================

5-Fold CV R² scores: [ 0.105 -0.147 -0.005 -0.06  -1.31 ]
Mean CV R²:  -0.2835
Std CV R²:   0.5199
```

In [19]:

```
print("="*45)
print("🏆 Best Model Summary")
print("="*45)
print("Model:  Random Forest v3")
print("Data:   Outliers removed (price <= $441)")
print("Target: Log-transformed price")
print(f"MAE:    $55.56  (was $74.70 originally)")
print(f"RMSE:   $74.84  (was $126.17 originally)")
print(f"R²:     0.198   (was 0.273 originally)")
print("="*45)
print("\n📉 MAE improved by 26% from the original model")
print("📉 RMSE improved by 41% from the original model")
```

```
============================================
🏆 Best Model Summary
============================================
Model:  Random Forest v3
Data:   Outliers removed (price <= $441)
Target: Log-transformed price
MAE:    $55.56  (was $74.70 originally)
RMSE:   $74.84  (was $126.17 originally)
R²:     0.198   (was 0.273 originally)
============================================

📉 MAE improved by 26% from the original model
📉 RMSE improved by 41% from the original model
```

In [ ]: